LESSON 12 · SESSION 1 · EXPLORE RATIO CONCEPTS

Four Test Tubes for Every One Student

E L A + M A T H · W O R D S T U D Y

Mr. H Math · Chandler Park Academy · Day 18

C C S S
6.RP.A.1
L.6.4
SMP 2,3,6

WRITE YOUR NAME — one letter per box, straight and neat

Date: _____ **Score: ___ / 10**

For five days every question had one answer: how much space does this shape cover? Today we stop measuring one thing and start comparing two. "4 test tubes for every 1 student" does not tell you how many students are in the room — it tells you how the two quantities are related. Every blank you fill in unlocks a word from today's vocabulary, so read closely.

DO NOW · SILENT START — 3 min

On the board: **12 and 4** described a shape yesterday. Today it says **"4 test tubes for every 1 student."** Write one way these are the same and one way they are different.

TURN & TALK — 2 min

Partner A reads the test-tube sentence out loud using *for every*. **Partner B** names the two quantities being compared — then switch.

Frames: "This compares _____ to _____." · "It does not tell me _____."

Section A · Guess the Word

Fill in the blanks to find the two words that signal a ratio relationship.

F O R [] V [] R Y

The SAME missing letter goes in both dashed boxes. Write it here: _____

There are 4 test tubes _____ 1 student.

A ratio compares two quantities. The words **for every** (or **for each**) say the comparison repeats for every group.

Section B · Meet the Ratio Family

Use the word bank to complete each sentence.

ratio · for every · quantity · order

1. A comparison of two quantities is called a _____.
 Example: 4 test tubes to 1 student.
2. Each of the two amounts a ratio compares is a _____.
 Example: test tubes is one; students is the other.
3. The words _____ tell you the comparison repeats for each group.
 Example: 6 tacos for every 3 guests — then 12 for 6 guests.
4. "5 tacos for every 2 guests" and "2 tacos for every 5 guests" are two different parties, so the _____ of the words sets the order of the numbers.
 Example: flip the words, flip the numbers.

Section C · Try It: Build the Groups

A circle is a student. A square is a test tube. Study the worked example, then build your own and write the ratio two ways.

WORKED EXAMPLE

4 test tubes for every 1 student

●□□□□ ●□□□□ ●□□□□

3 students → 3 × 4 = **12 test tubes**

students : test tubes = **3 : 12**
↑ the same relationship as **1 : 4**

YOUR TURN

3 test tubes for every 1 student, and there are **5 students**. Draw the groups in the space, then finish both lines.

5 students → 5 × 3 = _____ test tubes

students : test tubes = _____ : _____

REFINE · SUMMARY — A ratio is a comparison of two quantities, and the words **for every** are how you say one out loud. Saying the numbers is not enough — you have to name both quantities, in the order the words gave them. Next: writing that same ratio with the word **to** and with a **colon**, and finding the part-to-part and part-to-whole ratios hiding in one situation.

☆ Say it out loud: for every! ☆

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